Dr Kyle A Oman

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Institute for Computational Cosmology
Department of Physics & Astronomy, Durham University
South Road, Durham DH1 3LE, United Kingdom

Academic Qualifications

PhD University of Victoria, Astronomy — Aug 2017
Dissertation: "An explanation for the unexpected diversity of dwarf galaxy rotation curves"
Supervisor: Julio Navarro

MSc University of Waterloo, Physics — Aug 2013
Dissertation: "Probing the environmental dependence of star formation in satellite galaxies using orbital kinematics"
Supervisor: Michael Hudson

BSc University of Waterloo, Honours Physics (Astrophysics spec.) — Jun 2011
Graduated on Dean's Honour List
Dissertation: "An object-oriented halo finder"
Supervisor: Michael Balogh

Employment

Assoc. Professor & Royal Society Dorothy Hodgkin Fellow — Jul 2024 – present
Institute for Computational Cosmology &
Centre for Extragalactic Astronomy, Durham University

Asst. Professor & Royal Society Dorothy Hodgkin Fellow — Oct 2023 – Jun 2024
Institute for Computational Cosmology &
Centre for Extragalactic Astronomy, Durham University

Senior postdoctoral research associate — Jul 2022 – Sept 2023
Institute for Computational Cosmology, Durham University

Postdoctoral research associate — Oct 2019 – Jun 2022
Institute for Computational Cosmology, Durham University

Researcher (postdoctoral) — Oct 2017 – Sept 2019
Kapteyn Astronomical Institute, Rijksuniversiteit Groningen

Summer research internship — May 2010 – Aug 2010
Department of Physics & Astronomy, University of Waterloo

Summer research internship — May 2009 – Aug 2009
Department of Physics & Astronomy, University of Waterloo

Grants and Observing & Computing Time

Grants & research funding

RAS Summer undergraduate bursary (GBP 1 200) 2025
Title: “Dark matter in low-mass galaxies”
Role: Supervisor (PI)

Matariki Network of Universities Seed Fund (GBP 24 950) 2024 – 2026
Title: “The SWAN Universe: Simulations for Wallaby and the nearby Universe”
Role: co-I, Durham University lead

Durham STFC Impact Acceleration Account (GBP 17 156) 2024
Title: “Human mobility for natural disaster risk management with astrophysics techniques”
Role: co-PI (with A. Dunant)

Royal Society Dorothy Hodgkin Fellowship (GBP 1.5M) 2023 – 2031
Title: “Key dark matter particle properties from dwarf galaxy astrophysics”
Role: Fellow (PI)

Cosmology and Astroparticle Student & Postdoc Exchange Network Visitor (GBP 1 400) 2022
Title: “Structure and equilibrium of simulated dwarf galaxies”
Role: PI

Durham University Physics Department Developing Talent Award (GBP 4 000) 2022
Title: “The fragility of dwarf galaxies used as dark matter tracers”
Role: PI

Computing time

DiRAC 17th RAC call for proposals (168M cpu-hr) 2025 – 2028
Title: “Virgo I: The formation, evolution and clustering of galaxies”
Role: co-I

DiRAC 16th RAC call for proposals (274M cpu-hr) 2024 – 2027
Title: “Virgo I: The formation, evolution and clustering of galaxies”
Role: co-I

DiRAC-3 Phase 2 Director’s Discretionary Time (8.84M cpu-hr) 2023 – 2024
Title: “Simulated 21-cm survey of the Southern sky”
Role: PI

Observing time

VLA Semester 2026A (115.6 hrs, priority B) 2025
Title: “Adding Gas to the Story: HI-SAGA Survey of satellites around Milky Way Analogs”
Role: co-I (PI: Jones)

VLA Semester 2025A (46.7 hrs, priority C) 2024
Title: “Adding Gas to the Story: HI-SAGA Survey of satellites around Milky Way Analogs”
Role: co-I (PI: Jones)

MeerKAT Sept 2020 call (48 hrs, priority A) 2020
Title: "An HI perspective on galaxy evolution in Abell 2626 and its surroundings"
Role: co-I (PI: Deb)

Scholarships & Awards

Durham University Discretionary Award 2023
GBP 500, Institution-level, award by nomination

Durham University Discretionary Award 2022
GBP 1 000, Institution-level, award by nomination

RM Petrie Memorial Fellowship 2016 – 2017
CAD 4 750, Institution-level, award by nomination

University of Victoria President's Research Scholarship 2016
CAD 4 000, Institution-level, award by nomination

National Science and Engineering Research Council Canada Graduate Scholarship with Michael Smith Foreign Study Supplement 2015 – 2016
CAD 76 000, National-level, top Canadian graduate research scholarship
Title: "Remaining challenges to the standard model of cosmology: a solution to the cusp-core problem"

University of Victoria Graduate Award 2014 – 2015
CAD 6 000, Institution-level, award by nomination

Nora and Mark DeGoutière Memorial Scholarship 2014
CAD 11 250, Institution-level, award by nomination

University of Victoria Fellowship 2013
CAD 15 000, Institution-level, award by nomination

Queen Elizabeth II Graduate Scholarship in Science and Technology 2012
CAD 15 000, Provincial-level, research scholarship
Title: "Deconstructing Star Formation Histories from Orbits in N-Body Simulations"

National Science and Engineering Research Council Undergraduate Student Research Award 2009
CAD 4 500, National-level, summer research scholarship

Research supervision

Degree students as lead supervisor: 1 PhD, 9 Mphys/MSc, 1 BSc; as co-supervisor: 3 PhD, 2 BSc.

PhD:

1. PhD thesis supervisor (with F. Fragkoudi) for Diego Dado Oct 2024 – present
   Institute for Computational Cosmology, Durham University

2. PhD thesis co-supervisor (with F. Fragkoudi) for Martyna Winiarska — Oct 2024 – present
   Institute for Computational Cosmology, Durham University

3. Visiting PhD thesis co-supervisor (with A. Pontzen, C. Frenk) for Xu Zhao — Dec 2025 – present
   Institute for Computational Cosmology, Durham University

4. PhD thesis co-supervisor (with R. Massey, C. Frenk) for Ellen Sirks — Jan 2020 – Oct 2022
   Institute for Computational Cosmology, Durham University

Postdoctoral:

5. Postdoctoral researcher Dr Katherine Harborne — Feb 2025 – present
   Institute for Computational Cosmology, Durham University

MSc/MPhys:

6. MPhys thesis supervisor (with A. Pontzen) for Jemma Evelyn — Oct 2025 – present
   Institute for Computational Cosmology, Durham University

7. MPhys thesis supervisor (with M. Swinbank) for Brad Makinson — Oct 2024 – Apr 2025
   Institute for Computational Cosmology, Durham University
   "Seeing dark matter with MeerKAT:MHONGOOSE"

8. MPhys thesis supervisor (with M. Swinbank) for Jack Carter — Oct 2024 – Apr 2025
   Institute for Computational Cosmology, Durham University
   "Seeing dark matter with MeerKAT:MHONGOOSE"

9. MPhys thesis supervisor (with M. Swinbank) for Lauryn Tapper — Oct 2022 – Apr 2023
   Institute for Computational Cosmology, Durham University
   "A dynamic analysis of the gas within simulated galaxies"

10. MPhys thesis supervisor (with A. Fattahi) for Alex Cooke — Oct 2021 – Apr 2022
    Institute for Computational Cosmology, Durham University
    "The lives and deaths of faint satellite galaxies"

11. MPhys thesis supervisor (with C. Frenk) for Richard Brooks — Oct 2021 – Apr 2022
    Institute for Computational Cosmology, Durham University
    "A path to revealing the nature of dark matter by neutral gas"

12. MPhys thesis supervisor (with C. Frenk) for Finn Roper — Jun 2020 – Apr 2021
    Institute for Computational Cosmology, Durham University
    "The effect of baryon feedback on simulated dark matter distributions"

13. MSc thesis supervisor (with M. Verheijen) for Anatolii Zadvornyi — Sept 2019 – Jul 2021
    Kapteyn Institute, Rijksuniversiteit Groningen
    "Star formation suppression, gas consumption and stripping in cluster satellites"

14. MSc thesis supervisor (with S. Trager) for Amit Upadhyay — Sept 2018 – Aug 2020
    Kapteyn Institute, Rijksuniversiteit Groningen
    "Star formation histories of Coma Cluster galaxies matched to simulated orbits hints at quenching around first pericentre"

BSc:

15. BSc thesis supervisor (with M. Verheijen) for Anatolii Zadvornyi — Jan 2019 – May 2019
    Kapteyn Astronomical Institute, Rijksuniversiteit Groningen
    “Enviromental star formation suppression in galaxy clusters”
16. BSc thesis co-supervisor (with J. Navarro, A. Fattahi) for James Lane — Sept 2016 – Apr 2017
    Department of Physics & Astronomy, University of Victoria
    “The mysterious progenitor of the Ophiuchus stream”
17. BSc thesis co-supervisor (with J. Navarro) for Patrick McManus — Sept 2016 – Apr 2017
    Department of Physics & Astronomy, University of Victoria
    “The effect of an impulsive gravitational perturbation on a dark matter halo”

Summer & visiting students:

18. Visiting PhD student Julen Expósito-Márquez — Sept 2025 – Dec 2025
    Institute for Computational Cosmology, Durham University
19. Summer undergraduate research project supervisor for Helena Chase — June 2024 – Aug 2024
    Institute for Computational Cosmology, Durham University
20. Summer undergraduate research project supervisor for Brad Makinson — June 2024 – July 2024
    Institute for Computational Cosmology, Durham University
21. Summer research placement supervisor for Loretta Lanigan — July 2024 – Aug 2024
    Institute for Computational Cosmology, Durham University
22. Summer undergraduate research project supervisor for Mary Carstairs — July 2024 – Aug 2024
    Institute for Computational Cosmology, Durham University
23. Summer Nuffield Foundation Research Placement Host for Matilda Hunnisett — Aug 2023
    Institute for Computational Cosmology, Durham University
24. Summer Nuffield Foundation Research Placement Host for Piotr Stelmaszczyk — Aug 2023
    Institute for Computational Cosmology, Durham University
25. Summer undergraduate research project supervisor for Eleanor Downing — Jul – Sept 2022
    Institute for Computational Cosmology, Durham University
    “The diverse perturbations affecting dwarf galaxies used as dark matter tracers”
26. Summer Nuffield Foundation Research Placement Host for Jared Turnbull — Aug 2021
    Institute for Computational Cosmology, Durham University
    “Visualisations of Simulated Galaxies”

Teaching Experience

Course development — May – Aug 2015

Produced new draft of laboratory manual for University of Victoria undergraduate introductory astronomy course, in use as of Fall 2016. Development of visualization software for use in same course.

Substitute lecturer

Lectures for courses "Introduction to astronomy; Cosmic history" (Feb-Mar 2023, instructor C. Frenk), "Introduction to galaxies" (Mar 2017, instructor J. Navarro).

Guest lecturer

Lectures for courses "Representation of time travel in popular culture" (May 2014, Dept. Fine Arts, instructor J. Threlfall), "Life in the Universe" (Mar 2015, Mar 2016, Dept. Physics & Astronomy, instructor J. Willis).

Teaching Assistantships – 11 appointments totalling 45 months 2010 – 2022

Variously: demonstration and grading for introductory physics and astronomy laboratory courses; guided problem solving sessions and grading for introductory engineering courses; grading for physics and astronomy courses; demonstration and grading for MSc scientific computing course.

Publications

I am an author of 75 refereed articles (4498 citations excl. astropy paper, H=37), of which 11 as the first author (1027 citations), and a further 26 in which I had a major role. An up-to-date list is available on NASA ADS. I have highlighted some of my best work with a ★. Papers led by students under my supervision are marked with a †.

Pre-prints

1. Ludlow, Proctor, Schaye, Huško, Forouhar Moreno, Obreschkow, Chaikin, Schaller, Ploeckinger, Benítez-Llambay, **Oman**, McGibbon, Trayford, Frenk and Richings (2026). *The evolution of the sizes and angular momentum content of galaxies in the COLIBRE simulations.* MNRAS, submitted. [arxiv:2603.26200]

2. † Chase, Dado, Harborne and **Oman** (2025). *Consequences of radially correlated rotation curves for galaxy mass models.* MNRAS, submitted. [arxiv:2511.23106]

Refereed publications in primary journals

3. Lagos, Schaye, Schaller, Obreschkow, Bahé, Benítez-Llambay, Chaikin, Correa, Davis, Frenk, Huško, Kaasinen, McGibbon, **Oman**, Ploeckinger, Richings, Trayford, Wang and Wright (2025). *Kennicutt-Schmidt relation of galaxies over 13 billion years in the COLIBRE hydrodynamical simulations.* MNRAS, accepted. [arxiv:2512.11309]

4. Benavides, Sales, Navarro, White, Frenk, **Oman** and Cole (2025). *The abundance of thin dwarf galaxies: a challenge for cosmological simulations.* The Open Journal of Astropyhysics 9: 62091. [arxiv:2512.11035]

5. † Merrow, **Oman** and Fattahi (2026). *The lives and death of faint satellite galaxies around M31.* MNRAS 548: stag558. [arxiv:2511.01977]

6. Ponomareva, Mancera-Piña, Glowacki, Desmond, Jarvis, Varasteanu, Yasin, Heywood, Maddox, Adams, Baes, Gebek, Kurapati, Maksymowicz-Maciata, **Oman**, Pan, Prandoni, Rajohnson, Ruffa and Spekkens (2026). *MIGHTEE-HI: mass models and dark matter properties.* MNRAS 548: stag531. [arxiv:2603.16992]

7. † Makinson, **Oman** and Swinbank (2026). *Multi-resolution kinematic modelling of nearby galaxies: a demonstration using MHONGOOSE observations.* MNRAS 548: stag326. [arxiv:2508.08841]
8. †★ Dado, **Oman**, Harborne, Fragkoudi, Schaller, Schaye, Benítez-Llambay, Chaikin, Frenk, Huško, Ploeckinger and Richings (2026). *Dynamical disequilibrium in dwarf galaxies: rethinking gas dynamics, rotation curves, and dark matter inference.* MNRAS 547: stag356. [arxiv:2512.11033]
9. Riley, Shipp, Simpson, Bieri, Fattahi, Brown, **Oman**, Fragkoudi, Gómez, Grand and Marinacci (2025). *Auriga Streams I: distrupting satellites surrounding Milky Way-mass haloes at multiple resolutions.* MNRAS 542: 2443. [arxiv:2410.09144]
10. Namumba, Ianjamasimanana, Koribalski, Bosma, Athanassoula, Carignan, Jozsa, Kamphuis, Deane, Sikhosana, Verdes-Montenegro, Sorgho, Ndaliso, Amram, Brinks, Chemin, Combes, de Blok, Deg, English, Healy, Kurapati, Marasco, McGaugh, **Oman**, Spekkens, Veronese and Wong (2025). *Investigating the HI distribution and kinematics of ESO444-G084 and [KKS2000]23: New insights from the MHONGOOSE survey.* A&A 699: A372. [arxiv:2506.04101]
11. Hank, Verheijen, Blyth, Davé, **Oman**, Deg and Glowacki (2025). *HI asymmetries in spatially resolved SIMBA galaxies.* MNRAS 540: 3047. [arxiv:2506.15827]
12. † Visser-Zadvornyi, Carstairs, **Oman** and Verheijen (2025). *Star formation and stellar & AGN feedback in the absence of accretion, not gas stripping, set the quenching timescale in satellite galaxies.* MNRAS 540: 1730. [arxiv:2503.15183]
13. O'Beirne, Staveley-Smith, Kilborn, Wong, Westmeier, Cluver, Bekki, Deg, Dénes, For, Lee-Waddell, Murugeshan, **Oman**, Rhee, Shen and Taylor (2025). *WALLABY pilot survey: properties of HI-selected dark sources and low surface brightness galaxies".* PASA 42: 870. [arxiv:2505.04299]
14. Marasco, de Blok, Maccagni, Fraternali, **Oman**, Oosterloo, Combes, McGaugh, Kamphuis, Spekkens, Kleiner, Veronese, Amram, Chemin and Brinks (2025). *HI within and around observed and simulated galaxy discs – Comparing MeerKAT observations with mock data from TNG50 and FIRE-2*. A&A 697: 86. [arxiv:2503.03818]
15. Perron-Cormier, Deg, Spekkens, Richardson, Glowacki, Verheijen, Hank, Blyth, **Oman**, Dénes, Rhee, Elagali, Shen, Raja, Lee-Waddell and Westmeier (2025). *WALLABY pilot survey & Asymba: Comparing HI detection asymmetries to the SIMBA simulation.* AJ 169: 114. [arxiv:2501.09547]
16. Amvrosiadis, Lange, Nightingale, He, Frenk, **Oman**, Smail, Swinbank, Fragkoudi, Gadotti, Cole, Borsato, Robertson, Massey, Cao and Li (2025). *The onset of bar formation in a massive galaxy at z~3.8.* MNRAS 537: 1163. [arxiv:2404.01918]
17. Maccagni, de Blok, Mancera Piña, Ragusa, Iodice, Spavone, McGaugh, **Oman**, Oosterloo, Koribalski, Adams, Amram, Bosma, Bigiel, Brinks, Chemin, Combes, Gibson, Healy, Holwerda, Józsa, Kamphuis, Kleiner, Kurapati, Marasco, Spekkens, Veronese, Walter, Zabel and Zijlstra (2024). *MHONGOOSE discovery of a gas-rich low-surface brightness galaxy in the Dorado Group*. A&A 690: 69. [arxiv:2405.17000]

18. De Blok, Healy, Maccagni, Pisano, Bosma, English, Jarrett, Marasco, Meurer, Veronese, Bigiel, Chemin, Fraternali, Holwerfa, Kamphuis, Klöckner, Kleiner, Leroy, Mogotsi, **Oman**, Schinnerer, Verdes-Montenegro, Westmeier, Wong, Zabel, Amram, Carignan, Combes, Brinks, Dettmar, Gibson, Jozsa, Koribalski, McGaugh, Oosterloo, Spekkens, Schröder, Adams, Athanassoula, Bershady, Beswick, Blyth, Elson, Frank, Heald, Henning, Kurapati, Loubser, Lucero, Meyer, Namumba, Oh, Sardone, Sheth, Smith, Sorgho, Walter, Williams, Woudt and Zijlstra (2024). *MHONGOOSE – A MeerKAT Nearby Galaxy HI Survey.* A&A 688: A109. [arxiv:2404.01774]

19. **Oman**, Frenk, Crain, Lovell and Pfeffer (2024). *A warm dark matter cosmogony may yield more low-mass galaxy detections in 21-cm surveys than a cold dark matter one.* MNRAS 533: 67. [arxiv:2401.11878]

20. Brown, Fattahi, McCarthy, Font, **Oman** and Riley (2024). *ARTEMIS emulator: exploring the effect of cosmology and galaxy formation physics on Milky Way-mass haloes and their satellites*. MNRAS 532: 1223. [arxiv:2403.11692]

21. **Oman** and Riley (2024). *An overlooked source of uncertainty in the mass of the Milky Way.* MNRAS Letters 532: 48. [arxiv:2404.03726]

22. Jones, Sand, Karunakaran, Spekkens, **Oman**, Bennet, Besla, Crnojević, Cuillandre, Fielder, Gwyn and Mutlu-Pakdil (2023). *Gas and star formation in satellites of Milky Way analogs.* ApJ 966: 93. [arxiv:2311.02152]

23. † Sirks, Harvey, Massey, **Oman**, Robertson, Frenk, Everett, Gill and McCleary (2023). *Hydrodynamical simulations of merging galaxy clusters: giant dark matter particle colliders, powered by gravity.* MNRAS 530: 3160. [arxiv:2405.00140]

24. Puglisi, Dudzevičiūtė, Swinbank, Gillman, Tiley, Cirasuolo, Cortese, Glazebrook, Harrison, Ibar, Molina, Obreschkow, **Oman**, Schaller, Shankar and Sharples (2023). *KURVS: The outer rotation curve shapes and dark matter fractions of z~1.5 star-forming galaxies.* MNRAS 524: 2814. [arxiv:2305.04382]

25. † Brooks, **Oman** and Frenk (2023). *The North-South asymmetry of the ALFALFA HI velocity width function*. MNRAS 522: 4043. [arxiv:2211.08092]

26. † Downing and **Oman** (2023). *The many reasons that the rotation curves of low-mass galaxies can fail as tracers of their matter distributions.* MNRAS 522: 3318. [arxiv:2301.05242]

27. Reeves, Hudson and **Oman** (2023). *Constraining satellite quenching timescales in galaxy clusters by forward-modelling stellar ages and quiescent fractions in projected phase space.* MNRAS 522: 1779. [arxiv:2211.09145]

28. †★ Roper, **Oman**, Frenk, Benítez-Llambay, Navarro and Santos-Santos (2023). *The diversity of rotation curves of simulated galaxies with cusps and cores*. MNRAS 521: 1316. [arxiv:2203.16652]

29. Astropy Collaboration: Price-Whelan, Lim, Earl, Starkman, Bradley, Shupe, Patil, Corrales, Brasseur, Nöthe, Donath, Tollerud, Morris, Ginsburg, Vaherm Weaver, Tocknell, Jamieson, van Kerkwijk, Robitaille, Merry, Bachetti, and paper authors: Aldcroft, Alvaro-Montes, Archibald, Bódi, Bapat, Barentsen, Bazán, Biswas, Boquien, Burke, Cara, Cara, Conroy, Conseil, Craig, Cross, Cruz, D'Eugenio, Dencheva, Devillepoix, Dietrich, Eigenbrot, Erben, Ferreira, Foreman-Mackey, Fox, Freij, Garg, Geda, Glattly, Gondhalekar, Gordon, Grant, Greenfield, Groener, Guest, Gurovich, Handberg, Hart, Hatfield-Dodds, Homeier, Hosseinzadeh, Jenness, Jones, Joseph, Kalmback, Karamehmetoglu, Kałuszyński, Kelley, Kern, Kerzendorf, Kock, Kulumani, Lee, Ly, Ma, MacBride, Maljaars, Muny, Murphy, Norman, O'Steen, **Oman**, Pacifici, Pascual, Pascual-Granado, Patil, Perren, Pickering, Rastogi, Roulston, Ryan, Rykoff, Sabater, Sakurikar, Salgado, Sanghi, Saunders, Savchenko, Schwardt, Seifert-Eckert, Shih, Jain, Shukla, Sick, Simpson, Singanamalla, Singer, Singhal, Sinha, Sipőcz, Spitler, Stansby, Streicher, Šumak, Swinbank, Taranu, Tewary, Tremblay, de Val-Borro, van Kooten, Vasović, Verma, de Miranda Cardoso, Williams, Wilson, Winkel, Wood-Vasey, Xue, Yoachim, Zhang and Zonca (2022). *The Astropy project: Sustaining and growing a community-oriented open-source project and the latest major release (v5.0) of the core package.* ApJ 935: 167. [arxiv:2206.14220]
30. Bilimogga, **Oman**, Verheijen and van der Hulst (2021). *Using EAGLE simulations to study the effect of observational constraints on determination of HI asymmetries in galaxies*. MNRAS 513: 5310. [arxiv:2205.00675]
31. Mancera-Piña, Fraternali, Oosterloo, Adams, **Oman** and Leisman (2022). *No need for dark matter: resolved kinematics of the ultra-diffuse galaxy AGC 114905.* MNRAS 512: 3230. [arxiv:2112.00017]
32. † Sirks, **Oman**, Robertson, Massey and Frenk (2021). *The effects of self-interacting dark matter on the stripping of galaxies that fall into clusters*. MNRAS 511: 5927. [arxiv:2109.03257]
33. **Oman** (2022). *The ALFALFA HI velocity width function.* MNRAS 509: 3268. [arxiv:2108.08856]
34. Karunakaran, Spekkens, **Oman**, Simpson, Fattahi, Sand, Bennet, Crnojević, Frenk, Gómez, Grand, Jones, Marinacci, Mutlu-Pakdil, Navarro and Zaritsky (2021). *Satellites around Milky Way analogs: Tension in the number and fraction of quiescent satellites seen in observations versus simulations.* ApJ 916: 19. [arxiv:2105.09321]
35. † Upadhyay, **Oman** and Trager (2021). *Star formation histories of Coma Cluster galaxies matched to simulated orbits hint at quenching around first pericenter.* A&A 652: A16. [arxiv:2104.04388]
36. Brouwer, **Oman**, Valentijn, Bilicki, Heymans, Hoekstra, Napolitano, Roy, Tortora, Wright, Asgari, van den Busch, Dvornik, Erben, Giblin, Graham, Hildebrandt, Hopkins, Kannawadi, Kuijken, Liske, Shan, Tröster and Visser (2021). *The weak lensing radial acceleration relation: Constraining modified gravity and cold dark matter theories with KiDS-1000.* A&A 650: A113. [arxiv:2106.11677]
37. Board, Bozorgnia, Strigari, Grand, Fattahi, Frenk, Marinacci, Navarro and **Oman** (2021). *Velocity-dependent J-factors for annihilation radiation from cosmological simulations.* JCAP 2021-04: 70. [arxiv:2101.06284]

38. ★ **Oman**, Bahé, Healy, Hess, Hudson and Verheijen (2021). *A homogeneous measurement of the delay between the onsets of gas stripping and star formation quenching in satellite galaxies of groups and clusters.* MNRAS 501: 5073. [arxiv:2009.00667]
39. Deason, **Oman**, Fattahi, Schaller, Jauzac, Zhang, Montes, Bahé, Dalla Vecchia, Kay and Evans (2020). *Stellar splashback: the edge of the intracluster light.* MNRAS 500: 4181. [arxiv:2010.02937]
40. Genina, Read, Frenk, Cole, Benítez-Llambay, Ludlow, Navarro, **Oman** and Robertson (2020). *To beta or not to beta: can higher-order Jeans analysis break the mass-anisotropy degeneracy in simulated dwarfs?* MNRAS 498: 144. [arxiv:1911.09124]
41. Deason, Fattahi, Frenk, Grand, **Oman**, Garrison-Kimmel, Simpson and Navarro (2020). *The edge of the Galaxy.* MNRAS 496: 3929. [arxiv:2002.09497]
42. Mancera-Piña, Fraternali, **Oman**, Adams, Bacchini, Marasco, Oosterloo, Pezzulli, Posti, Leisman, Cannon, di Teodoro, Gault, Haynes, Reiter, Rhode, Salzer and Smith (2020). *Robust HI kinematics of gas-rich ultra-diffuse galaxies: hints of a weak-feedback formation scenario.* MNRAS 495: 3636. [arxiv:2004.14392]
43. Marasco, Posti, **Oman**, Famaey, Cresci and Fraternali (2020). *Massive disc galaxies in cosmological hydrodynamical simulations are too dark matter-dominated.* A&A 883: L33. [arxiv:2005.01724]
44. ★ Santos-Santos, Navarro, Robertson, Benítez-Llambay, **Oman**, Lovell, Frenk, Ludlow, Fattahi and Ritz (2020). *Baryonic clues to the puzzling diversity of dwarf galaxy rotation curves.* MNRAS 495: 58. [arxiv:1911.09116]
45. Cautun, Benítez-Llambay, Deason, Frenk, Fattahi, Gómez, Grand, **Oman**, Navarro and Simpson (2020). *The Milky Way total mass profile as inferred from Gaia DR2.* MNRAS 494: 4291. [arxiv:1911.04557]
46. Richings, Frenk, Jenkins, Robertson, Fattahi, Grand, Navarro, Pakmor, Gómez, Marinacci and **Oman** (2020). *Subhalo destruction in the APOSTLE and Auriga simulations.* MNRAS 492: 5780. [arxiv:1811.12437]
47. † Lane, Navarro, Fattahi, **Oman** and Bovy (2020). *The Ophiuchus stream progenitor: a new type of globular cluster and its possible Sagittarius connection.* MNRAS 492: 4164. [arxiv:1905.12633]
48. Chauhan, Lagos, Obreschkow, Power, **Oman** and Elahi (2019). *The HI velocity function: a test of cosmology or baryon physics?* MNRAS 488: 5898. [arxiv:1906.06130]
49. Genina, Frenk, Benítez-Llambay, Cole, Navarro, **Oman** and Fattahi (2019). *The distinct stellar metallicity populations of simulated Local Group dwarfs.* MNRAS 488: 2312. [arxiv:1812.04839]
50. Bose, Frenk, Jenkins, Fattahi, Gomez, Grand, Marinacci, Navarro, **Oman**, Pakmor, Schaye, Simpson and Springel (2019). *No cores in dark matter-dominated dwarf galaxies with bursty star formation histories.* MNRAS, 486: 4790. [arxiv:1810.03635]

51. Owers, Hudson, **Oman**, Bland-Hawthron, Brough, Bryant, Cortese, Couch, Croom, van de Sande, Federrath, Groves, Hopkins, Lawrence, Lorente, McDermid, Medling, Richards, Scott, Taranu, Welker and Yi (2019). *The SAMI Galaxy Survey: Quenching of star formation in clusters I. Transition galaxies.* ApJ 873: 52. [arxiv:1901.08185]

52. ★ **Oman**, Marasco, Navarro, Frenk, Schaye and Benítez-Llambay (2019). *Non-circular motions and the diversity of dwarf galaxy rotation curves.* MNRAS 482: 821. [arxiv:1706.07478]

53. Digby, Navarro, Fattahi, Simpson, **Oman**, Gomez, Frenk, Grand and Pakmor (2018). *The star formation histories of dwarf galaxies in Local Group cosmological simulations.* MNRAS 485: 5423. [arxiv:1812.04839]

54. Mancera-Piña, Fraternali, Adams, Marasco, Oosterloo, **Oman**, Leisman, di Teodoro, Posti, Battipaglia, Cannon, Gault, Haynes, Janowiecki, McAllan, Pagel, Reiter, Rhode, Salzer and Smith (2019). *Off the baryonic Tully-Fisher relation: a population of baryon-dominated ultra-diffuse galaxies.* ApJL 883: 33. [arxiv:1909.01363]

55. Fattahi, Navarro, Frenk, **Oman**, Sawala and Schaller (2018). *Tidal stripping and the structure of dwarf galaxies in the Local Group.* MNRAS 476: 3816. [arxiv:1707.03898]

56. Navarro, Yozin, Loewen, Benítez-Llambay, Fattahi, Frenk, **Oman**, Schaye and Theuns (2018). *The innate origin of radial and vertical gradients in a simulated galaxy disc.* MNRAS 476: 3648. [arxiv:1709.01040]

57. Marasco, **Oman**, Navarro, Frenk and Oosterloo (2018). *Bars in dark-matter-dominated dwarf galaxy discs.* MNRAS 476: 2168. [arxiv:1711.09914]

58. Genina, Benítez-Llambay, Frenk, Cole, Fattahi, Navarro, **Oman**, Sawala and Theuns (2018). *The core-cusp problem: a matter of perspective.* MNRAS 474: 1398. [arxiv:1707.06303]

59. Navarro, Benítez-Llambay, Fattahi, Frenk, Ludlow, **Oman**, Schaller and Theuns (2017). *The origin of the mass discrepancy-acceleration relation in ΛCDM.* MNRAS 471: 1841. [arxiv:1612.06329]

60. Campbell, Frenk, Jenkins, Eke, Navarro, Sawala, Schaller, Fattahi, **Oman** and Theuns (2017). *Knowing the unknowns: uncertainties in simple estimators of dynamical masses.* MNRAS 469: 2335. [arxiv:1603.04443]

61. Wang, Fattahi, Cooper, Sawala, Strigari, Frenk, Navarro, **Oman** and Schaller (2017). *Tidal features of classical Milky Way satellites in a Λ cold dark matter universe.* MNRAS 468: 4887. [arxiv:1611.00778]

62. Sawala, Pihajoki, Johansson, Frenk, Navarro, **Oman** and White (2017). *Shaken and stirred: The Milky Way's dark substructures.* MNRAS 467: 4383. [arxiv:1609.01718]

63. Ludlow, Benítez-Llambay, Schaller, Theuns, Frenk, Bower, Schaye, Crain, Navarro, Fattahi and **Oman** (2017). *The Mass-discrepancy acceleration relation: a natural outcome of galaxy formation in CDM halos.* Phys Rev Lett 118: 161103. [arxiv:1610.07663]

64. Benítez-Llambay, Navarro, Frenk, Sawala, **Oman**, Fattahi, Schaller, Schaye, Crain and Theuns (2017). *The properties of “dark” LCDM haloes in the Local Group.* MNRAS 465: 3913. [arxiv:1609.01301]

65. Starkenburg, **Oman**, Navarro, Crain, Fattahi, Frenk, Sawala and Schaye (2017). *The oldest and most pristine stars in the APOSTLE Local Group simulations.* MNRAS 465: 2212. [arxiv:1609.05214]
66. Sales, Navarro, **Oman**, Fattahi, Ferrero, Abadi, Bower, Crain, Frenk, Sawala, Schaller, Schaye, Theuns and White (2016). *The low-mass end of the baryonic Tully-Fisher relation.* MNRAS 464: 2419. [arxiv:1602.02155]
67. **Oman** and Hudson (2016). *Satellite quenching timescales in clusters from projected phase space measurements matched to simulated orbits.* MNRAS, 463: 3083. [arxiv:1607.07934]
68. Schaller, Frenk, Fattahi, Navarro, **Oman** and Sawala (2016). *The low abundance and insignificance of dark discs in simulated Milky Way galaxies.* MNRAS 461: L56. [arxiv:1605.02770]
69. **Oman**, Navarro, Sales, Fattahi, Frenk, Sawala, Schaller and White (2016). *Missing dark matter in dwarf galaxies?* MNRAS 460: 3610. [arxiv:1601.01026]
70. Sawala, Frenk, Fattahi, Navarro, Bower, Crain, Dalla Vecchia, Furlong, Helly, Jenkins, **Oman**, Schaller, Schaye, Theuns, Trayford and White (2016). *The APOSTLE simulations: solutions to the Local Group's cosmic puzzles.* MNRAS 457: 1931. [arxiv:1511.01098]
71. Fattahi, Navarro, Sawala, Frenk, **Oman**, Crain, Furlong, Schaller, Schaye, Theuns and Jenkins (2016). *The APOSTLE project: Local Group kinematic mass constraints and simulation candidate selection.* MNRAS 457: 844. [arxiv:1507.03643]
72. ★ **Oman**, Navarro, Fattahi, Frenk, Sawala, White, Bower, Crain, Furlong, Schaller, Schaye and Theuns (2015). *The unexpected diversity of dwarf galaxy rotation curves.* MNRAS. 452: 3650. [arxiv:1504.01437]
73. Taranu, Hudson, Balogh, Smith, Power, **Oman**, Krane (2014). *Quenching star formation in cluster galaxies.* MNRAS 440: 1934. [arxiv:1211.3411]
74. **Oman**, Hudson and Behroozi (2013). *Disentangling satellite galaxy populations using orbit tracking in simulations.* MNRAS 431: 2307. [arxiv:1301.6757]

Other refereed publications

75. **Oman** (2025). *SWIFTGalaxy: a python package to work with particle groups from SWIFT simulations.* JOSS 10: 9278. [arxiv:2510.22328]
76. **Oman** (2024). *MARTINI: Mock Array Radio Telescope Interferometry of the Neutral ISM.* JOSS 9: 6860. [arxiv:2406.05574]
77. **Oman**, Starkenburg and Navarro (2018). *The "building blocks" of stellar haloes.* Galaxies 5: 33. [arxiv:1708.00929]

Non-refereed publications & software

78. **Oman** (2025). *SWIFTGalaxy: Galaxy particle analyzer.* Astrophysics Source Code Library 2505.002.
79. **Oman** and Downing (2025). Low-mass galaxy rotation curves that fail as dynamical mass tracers. Dynamical masses of Local Group galaxies, Proceedings of the IAU Symposium 379: 329.
80. Read, Steger, Walker, Genina, Frenk, Cole, Benítez-Llambay, Ludlow, Navarro, **Oman**, Robertson, Collins, Ibata, Rich, Martin, Peñarrubia. Chapman, Tollerud and Weisz (2023). *GravSphere: Jeans modelling code.* Astrophysics Source Code Library 2312.009.

81. **Oman**, Brouwer, Ludlow and Navarro (2020). *Observational constraints on the slope of the radial acceleration relation at low accelerations.* [arxiv:2006.06700]
82. **Oman** (2019). *MARTINI: Mock spatially resolved spectral line observations of simulated galaxies.* Astrophysics Source Code Library 1911.005.
83. **Oman** (2017). *The APOSTLE simulations: Rotation curves derived from synthetic 21-cm observations.* Rediscovering our Galaxy, Proceedings of the IAU Symposium 334: 213. [arXiv: 1712.02562]
84. Fattahi, Navarro, Sawala, Frenk, Sales, **Oman**, Schaller and Wang (2016). *The cold dark matter content of Galactic dwarf spheroidals: no cores, no failures, no problem.* [arxiv:1607.06479]
85. Sawala, Frenk, Fattahi, Navarro, Bower, Crain, Dalla Vecchia, Furlong, Helly, Jenkins, **Oman**, Scahller, Schaye, Theuns, Trayford and White (2014). *Local Group galaxies emerge from the dark.* [arxiv:1412.2748]

Conferences and Workshops

Contributed talk at “Royal Society Meeting of Minds 2026” — Feb 2026
Royal Society, London, UK

Contributed talks at “SWIFT users & developers meeting 2025” — Oct 2025
Leiden University, Leiden, The Netherlands

Contributed talk at “Pathfinder HI Survey Coordination Committee workshop” — Sept 2025
Cagliari, Italy

SOC at “AstroDat” — Sept 2025
Durham University, Durham, UK

Contributed talk at “NAM2025/Galaxy formation simulations at the frontier” — July 2025
Durham University, Durham, UK

SOC at “SWAN Universe workshop” — Dec 2024
ICRAR UWA, Perth, Australia

Invited talk at IAUGA 2024/FM 9 “Measures of luminous and dark matter in galaxies across time” — Aug 2024
CTICC, Cape Town, South Africa

Invited plenary talk at IAUGA 2024/IAUS 392 “Neutral hydrogen in and around galaxies in the SKA era” — Aug 2024
CTICC, Cape Town, South Africa

SOC at “Small galaxies, cosmic questions II” — Aug 2024
Durham University, Durham, United Kingdom

Contributed talk at “Towards unified sub-grid prescriptions for galaxy modelling” — Sept 2023
Lorentz Centre, Leiden, The Netherlands

Contributed talk at “FiatLux” — Jun 2023
Centro Mariapoli, Castel Gandolfo, Italy

Contributed talk at “Pathfinder HI Survey Coordination Committee workshop” Mar 2023
University of Cape Town, Cape Town, South Africa

Poster at “IAUS 379: Dynamical masses of Local Group galaxies” Mar 2023
Telegrafenberg, Potsdam, Germany

Contributed talk at “Virgo meeting” Jul 2022
Max-Planck Institute for Astrophysics, Garching, Germany

Contributed talk at “EAS2022/SS8: Dwarf galaxies beyond the Local Group” Jun 2022
Valencia Conference Centre, Valencia, Spain

Poster at “EAS2022/S4: Satellite galaxies and tidal streams” Jun 2022
Valencia Conference Centre, Valencia, Spain

Invited talk at “EAS2022/SS5: Neutral hydrogen” Jun 2022
Valencia Conference Centre, Valencia, Spain

Contributed talk at “Durham Edinburgh Exchange” Jan 2022
Virtual meeting

Contributed talk at “Heraeus Seminar: Astrophysical windows on dark matter” Nov 2021
The Royal Society, London, United Kingdom

Contributed talk & 2 posters at “National Astronomy Meeting 2021” Jul 2021
Virtual meeting

SOC at “EAS2021/SS24: The role of nurture on the SF cycle in satellite galaxies” Jun 2021
Virtual meeting

Contributed talk at “Durham Edinburgh Exchange” Jan 2021
Virtual meeting

Contributed talk at “South American Dark Matter” Dec 2020
Virtual meeting

Contributed talk at “WALLABY Science Day” Nov 2020
Virtual meeting

Invited talk at “Pathfinder HI Survey Coordination Committee workshop” May 2020
Virtual meeting

Participant at “Virgo meeting” Jan 2020
Durham University, Durham, United Kingdom

Contributed talk at “Durham Edinburgh Exchange” Jan 2020
Durham University, Durham, United Kingdom

Contributed talk at “Galaxy evolution in a new era of HI surveys” Aug 2019
Munich Institute for Astro- and Particle Physics, Munich, Germany

Contributed talk at “Small galaxies, cosmic questions” Jul 2019
Durham University, Durham, United Kingdom

Contributed talk at "Computational cosmology" Dec 2018
Lorentz Centre, Leiden, The Netherlands

Participant at "Blaauw Workshop: Galaxy dynamics in the current era" Nov 2018
Kapteyn Institute, Groningen, The Netherlands

Contributed talk at "The HI/Story of the nearby Universe" Sept 2018
Kapteyn Institute/ASTRON, Groningen, The Netherlands

Contributed talk at "Tensions in the LCDM paradigm" May 2018
Mainz Institute for Theoretical Physics, Mainz, Germany

Poster at "The small-scale structure of cold(?) dark matter" Apr 2018
Kavli Institute for Theoretical Physics, Santa Barbara, USA

Contributed talk at "Virgo meeting" Dec 2017
Max-Planck Institute for Astrophysics, Garching, Germany

Invited talk at "IAUS 334: Rediscovering our Galaxy" Jul 2017
Telegrafenberg, Potsdam, Germany

Contributed talk at "On the origin of baryonic galaxy haloes" Mar 2017
Observatorio Astronómico de Quito, Galapagos Islands, Ecuador

Contributed talk at "Northwest astronomy meeting" Oct 2016
Western Washington University, Bellingham, USA

Contributed talk at "Dark matter in the Milky Way" May 2016
Johannes Gutenburg University, Mainz, Germany

Contributed talk at "Dark matter on the smallest scales" Apr 2016
University of Leiden, Leiden, Netherlands

Contributed talk at "Potsdam thinkshop XIII: Near field cosmology" Mar 2016
Innsbruck University Centre, Obergurgl, Austria

Participant at "HiPACC Summer School: AstroInformatics" Aug 2012
University of California at San Diego, San Diego, USA

Poster at "Star formation and gas reservoirs in nearby groups and clusters" Jul 2012
Union College, Schenectady, USA

Invited Colloquia & Seminars

Seminar: Oxford University Physics Department, UK Dec 2025

Seminar: University of Victoria Department of Physics & Astronomy, Canada Aug 2025

Seminar: NRC Herzberg Institute of Astrophysics, Victoria, Canada Aug 2025

Seminar: University of Ghent Department of Physics & Astronomy, Belgium Jun 2025

Colloquium: ICRAR, University of Western Australia, Australia Mar 2024

Colloquium: Cardiff University School of Physics & Astronomy, UK Nov 2023

Seminar: University of Edinburgh Institute for Astronomy, UK	Nov 2022
Seminar: University College London Cosmoparticle Initiative, UK	Mar 2022
Seminar: International Centre for Radio Astronomy Research, Australia	Aug 2021
Colloquium: Queen’s University Department of Physics and Astronomy, Canada	Jan 2021
Seminar: University of Nottingham School of Physics and Astronomy, UK	Dec 2020
Seminar: Oxford University Department of Physics, UK	Oct 2020
Seminar: Universidade Federal do Espírito Santo Department of Physics, Brazil	Jul 2020
Colloquium: University of Edinburgh Institute for Astronomy, UK	Feb 2020
Seminar: Deutsches Elektronen-Synchrotron DESY, Germany	Nov 2019
Seminar: University of Surrey Physics Department, UK	Sept 2017
Seminar: University of Cambridge Institute of Astronomy, UK	May 2016
Seminar: Astrophysics Institute Potsdam, Germany	Apr 2016
Seminar: University of Washington Physics & Astronomy Department, USA	Feb 2016
Seminar: University of Waterloo Physics & Astronomy Department, Canada	Dec 2014

Public Talks & Outreach

Public talk: Cleveland & Darlington Astronomical Society	Feb 2026
Public talk: Newcastle Astronomical Society	Sept 2024
Radio Astro Interviewee at IAU GA XXXIII	Aug 2024
Café Scientifique Durham speaker	May 2024
Durham Schools Science Festival demonstrator	Mar 2024
Street Cosmos at Blackhall demonstrator	Aug 2023
Ogden Centre for Fundamental Physics *Ogden@20* open day demonstrator	Nov 2022
Celebrate Science! Durham demonstrator	Oct 2022
Public talk: Sunderland Astronomical Society	Nov 2021
Nuffield Foundation research placement host for J. Turnbull	Aug 2021
Royal Society Summer Science Exhibition Workshop	Jul 2021
Public talk: Rato Bangala School Science Club, Nepal	Jul 2020
Durham ICC School visits	Jan 2020 – present
Celebrate Science! Durham demonstrator	Oct 2019
Meet the IAU astronomers! programme member	2019 – present

Public talk: RAS of Canada, Victoria chapter monthly meeting	Apr 2017
Public talk: Dominion Astrophysical Observatory summer star parties	Jul 2016
University of Victoria Student Radio CFUV "Beyond the Jargon" interviewee	Dec 2016
Public talk: SPACE: Students in Physics and Astronomy Communication Enrichment	Jul 2016
Public talk: SPACE: Students in Physics and Astronomy Communication Enrichment	Aug 2015
University of Victoria Observatory tours for various school, youth and general public groups	2015 – 2017
Contributor of >250 physics & astrophysics answers to public questions on the PhysicsSE Q&A platform (physics.stackexchange.com/users/11053/kyle-oman)	2013 – present

Professional Citizenship

Referee for MNRAS, ApJ, A&A, NatAs, PRL, JCAP, OJA, JOSS, Chinese Physics C	2016 – present
Durham Astronomy & Instrumentation internal mini-conference organiser	Jun 2022
Durham Physics Research Staff Consultative Committee member as co-chair: 2020 – 2023	2019 – present
Durham Astronomy PhD admissions interview panellist	2020
Durham Astronomy ICC postdoctoral representative to astronomy group	2020 – 2023
Durham Astronomy Friday seminar co-organizer	2019 – 2020
Kapteyn Institute Monday seminar co-organizer	2018 – 2019
Funding proposal reviews for UKRI/STFC (UK), SNF (Switzerland), FWO (Belgium), Conicyt (Chile)	2018
Observing proposal reviews for CFHT, Gemini.	2022 – present
University of Victoria Physics & Astronomy graduate student association vice-chair	2014 – 2015
University of Victoria astronomy group weekly discussion meeting chair	2014 – 2015
University of Victoria graduate student representative to department	2014 – 2015
University of Waterloo undergraduate student representative to department	2008 – 2011

Training & professional development

Presentation skills The Royal Society	Jan 2026
Leadership effectiveness	Oct 2024

The Royal Society & ICL Business School

Introduction to management — Sept 2024
The Royal Society & ICL Business School

Getting started with doctoral supervision — Jan 2024
Durham Centre for Academic Development (DCAD)

Research-informed public engagement — Jul 2023
DiRAC & Science Museums Group

Professional Memberships

WALLABY survey member (Technical Working Group 1) — 2020 – present
TWG 1 co-chair: 2024 – present

MIGHTEE-HI survey member — 2023 – present

MHONGOOSE survey member — 2020 – present

SKA Cosmology Science Working Group member — 2022 – present

SKA HI Galaxy Science Working Group member — 2021 – present

Virgo Consortium for Cosmological Supercomputer Simulations member — 2013 – present

Canadian Astronomical Society (CASCA) Ordinary Member — 2016 – 2019

Royal Netherlands Astronomical Society Member — 2017 – 2019

International Astronomical Union (IAU) Junior Member — 2018 – present

Fellow of the Royal Astronomical Society — 2024 – present

European Astronomical Society Member — 2025 – present

Public software & utilities

As lead author:

SWIFTGalaxy — 2022
github.com/SWIFTSIM/swiftgalaxy

MARTINI: Mock Array Radio Telescope Interferometry of the Neutral ISM — 2019
github.com/kyleaoman/martini
Code reviewed through PyOpenSci & published in JOSS

read_eagle (python-only version) — 2019
github.com/kyleaoman/pyread_eagle

eagleSQLTools (python3 version) — 2018
github.com/kyleaoman/eagleSqlTools

As contributor:

astropy, SWIFTSimIO (co-maintainer), unyt (part of yt), ^{3D}Barolo

Languages

English (native), French (native), Dutch (basic)

Citizenship

Canadian

References

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